WEEK 06 · SESSION 1

Validation, Cross-Validation and Tuning

Make model decisions with valid boundaries and decision-aware metrics.

DSACADEMY

Learning outcomes

01 Choose splits that reflect future use.

02 Tune without contaminating the test set.

03 Search hyperparameters within a reproducible budget.

The concept system

01

holdout

02

cross-validation

03

time splits

04

hyperparameters

A repeatable reasoning loop

01 **Define** → 02 **Transform** → 03 **Validate** → 04 **Communicate**

Design validation schemes for random and temporal datasets.

Guided lab

LAB BRIEF

Design validation schemes for random and temporal datasets.

workbook / tests / evidence

01 **Frame**

Define inputs, outputs, and success criteria.

02 **Build**

Implement the smallest correct workflow.

03 **Verify**

Test normal, boundary, and failure cases.

04 **Explain**

Record the evidence and remaining limits.

Portfolio deliverable

A validation protocol and reproducible tuning run.

QUALITY GATE

- Reproducible
- Tested
- Decision-relevant

WEEKLY ASSESSMENT

Evaluation Decision Memo

Validity 35%, metric reasoning 25%, error analysis 20%, uncertainty 10%, communication 10%.

SESSION CHECKPOINT

What should remain after the lesson?

CONCEPT	holdout + cross-validation
PRACTICE	Design validation schemes for random and temporal datasets.
EVIDENCE	A validation protocol and reproducible tuning run.